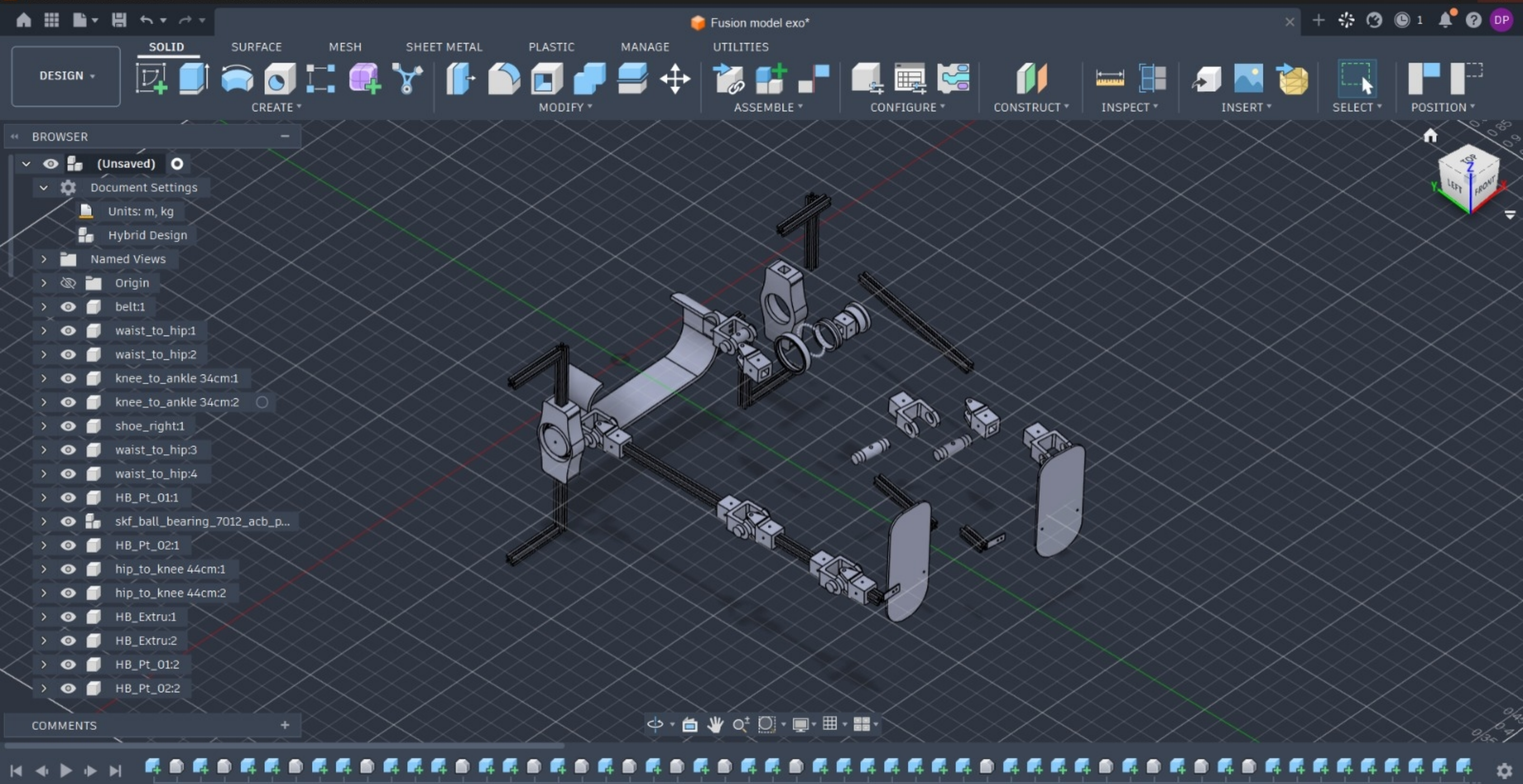






EXO-AUGMENT // VIRTUAL LAB

LOWER LIMB BIOMECHANICS & AAN REHABILITATION

● ACTUATOR BUS 48V [ONLINE]

|| Pause

SUBJECT ANTHROPOMETRICS

Weight (kg) Height (cm)
70 175

Thigh (cm) Shank (cm)
43.0 43.0

Foot Length (cm)
26.0

Auto-Fit Proportions (Winter BSP)

SEGMENT MASSES & INERTIA

Torso (HAT) Mass 47.46 kg
67.8% of 70kg (HAT)

Thigh Mass (each) 7 kg
10.0% of 70kg

Shank Mass (each) 3.25 kg
4.65% of 70kg

Foot Mass (each) 1.02 kg
1.45% of 70kg

Thigh Inertia (I_{zz}) 0.3774 kg·m²
Around Hip Axis

Shank Inertia (I_{zz}) 0.1678 kg·m²

Split View 3D Rig 2D Sagittal



3D PERSPECTIVE // ARTICULATED EXOSKELETON

Iso Sagittal Frontal Top



Stand-to-Sit (Descent) • Trunk Flexion

HIP JOINT 89.8°
Required: 64.4 N·m
Patient Muscle: 10.9 N·m
Exo Assist: 53.4 N·m

KNEE JOINT 89.8°
Required: 141.1 N·m
Patient Muscle: 24 N·m
Exo Assist: 100 N·m

ANKLE JOINT 9.6°
Required: 3.8 N·m
Patient Muscle: 0.6 N·m
Exo Assist: 3.1 N·m

Telemetry Graphs

Biomechanical Data Table

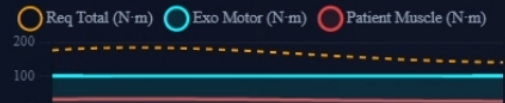
Split Analytics

Export Table (CSV)

JOINT ANGLES TRAJECTORY (°)



KNEE TORQUE SHARING (N·M)



ACTUATOR POWER



Standing Sway

Manual Jog

Simulation Speed: 0.9x

STROKE ASSIST-AS-NEEDED (AAN)

PATIENT STRENGTH
17%

EXO ASSISTANCE
83%

Patient Voluntary Muscle
Capability

Deficit
Compensated

CLINICAL PRESETS

Flaccid (0%)

Severe (15%)

Moderate (35%)

Mild (65%)

Active (100%)

ROBOTIC ACTUATORS & POWER

Knee Motor Elec Power: 55.3 W

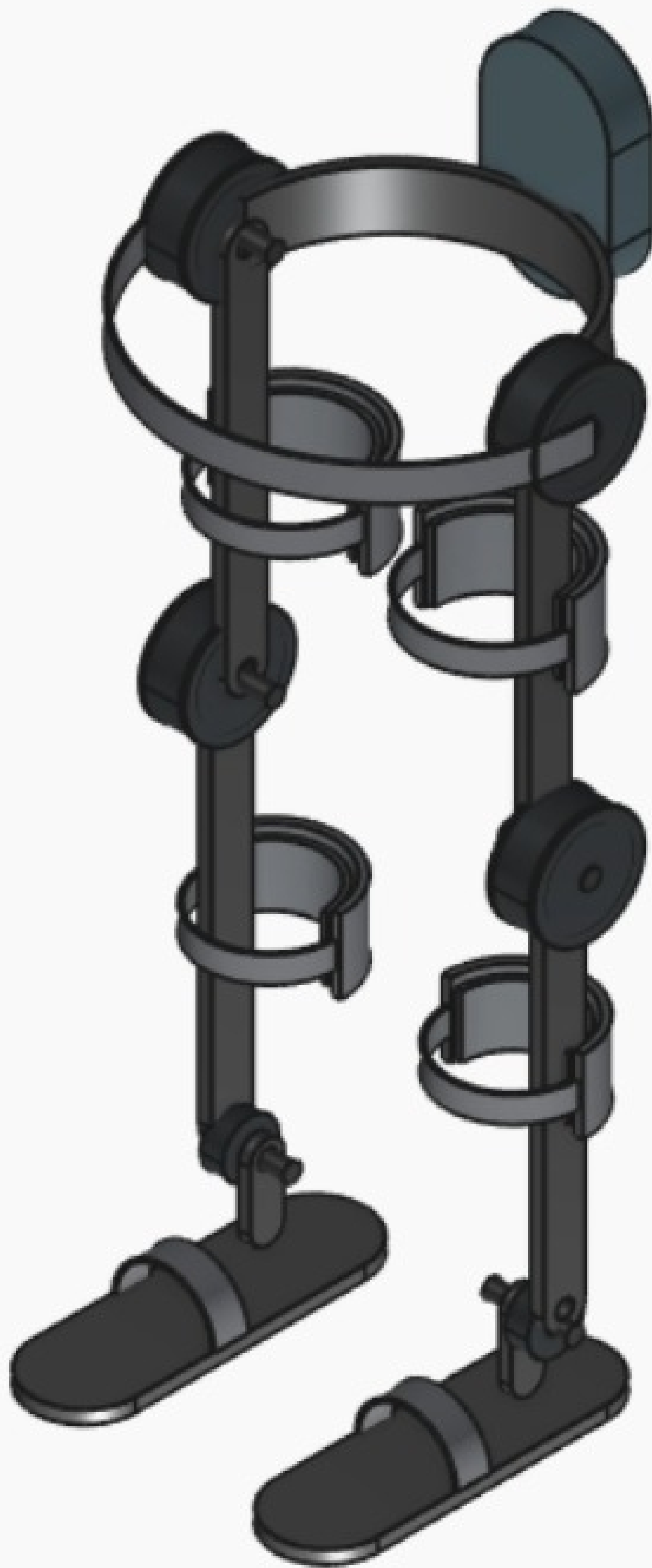
Knee Actuator Load: 100%

Total Power Draw: 89.3 W

48V DC Bus Current: 1.86 A

Battery Life (480Wh pack): 5.4 hrs

Export Biomechanical Report



Problem :- Patients after surgeries / stroke
or neurological conditions may have
weak or poorly controlled leg movement.

What we doing :-

An adaptable human augmentation
device - lower-limb exoskeleton.

Basic working

Sense $\rightarrow$ Estimate $\rightarrow$ Assist $\rightarrow$ Reevaluate $\rightarrow$
Adapt.

Sensors

- IMU $\Rightarrow$ leg movement / angle / velocity
- Pressure sensor $\Rightarrow$ foot contact / weight shift
- Encoder $\Rightarrow$ knee angle
- optional EMG $\Rightarrow$ muscle activity

+
safety mechanism.

Torque calculation :-

Gravity force ,

$$F = mg$$

Gravity torque , $\tau_g = mgr \sin \theta$

Angular velocity ,

$$\omega = \frac{d\theta}{dt}$$

Angular

acceleration , $\alpha = \frac{d\omega}{dt}$

Initial torque ,

$$\tau_i = I\alpha$$

Required torque ,

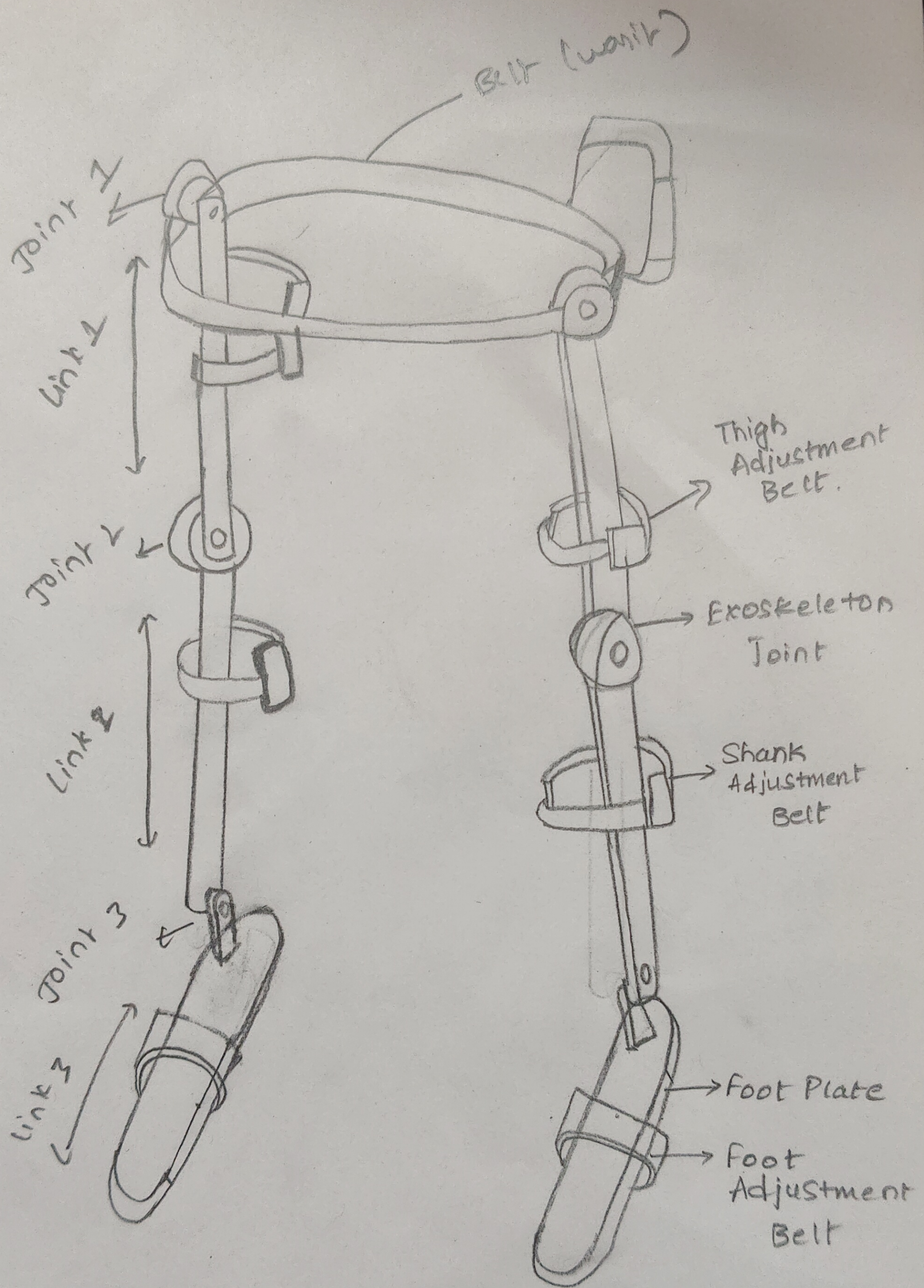
$$\tau_{\text{required}} = \tau_g + \tau_i + \tau_{\text{ext}}$$

for uniform rod ,

$$I = \frac{1}{3} mL^2$$

2) user contribution

$$\tau_{\text{user}}$$



LOWER LIMB EXOSKELETON
Human Augmentation

Assistance ;

$$\tau_{assist} = k_a \tau_{deficit}$$

$$\tau_{deficit} = \max(0, \tau_{req} - \tau_{user})$$

safety limits,

$$\tau_{safe} = \text{clip}(\tau_{assist} - \tau_{max}, +\tau_{max})$$

also limits,

- max torque
- torque rate
- knee angle
- Angular velocity
- sensor failures

Adaptation :

weak $\rightarrow$ assistance increases

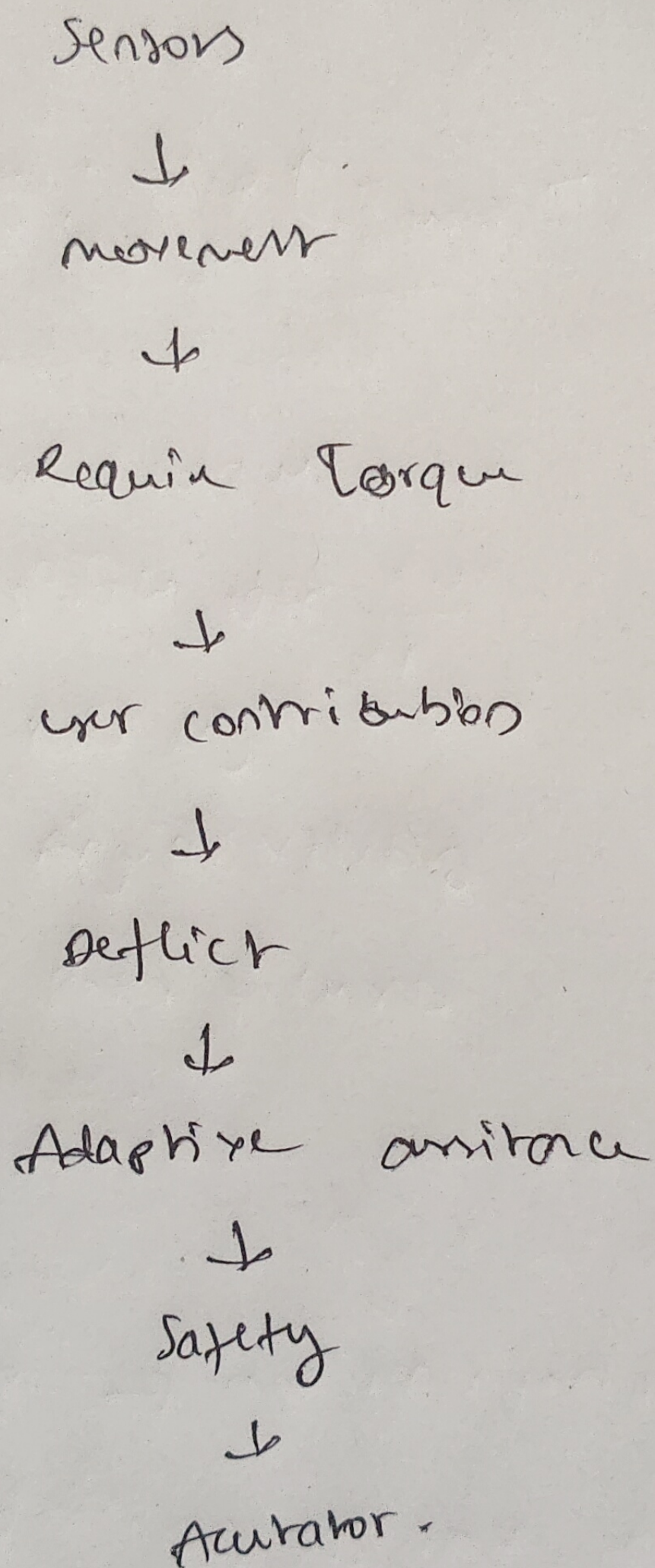
improving $\rightarrow$ assistance decreases

fatigue $\rightarrow$ temporary assistance
increases

sudden deterioration $\rightarrow$ protective
mode

$$\text{User contribution \%} = \frac{T_{\text{user}}}{T_{\text{required}}} \times 100$$

Control Flow



CAD model has been done in solidworks
and fusion. (Autocad Autodesk).